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Flak

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[54] **RACKET FRAME**

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[52] **U.S. Cl.** ..... **273/73 H**, 273/73 D, 273/73 C

[51] **Int. Cl.** **A63b 49/12**

[58] **Field of Search** ..... 273/73 R, 73 C, 73 D, 273/73 H

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[57] **ABSTRACT**

A tennis or the like game racket having a frame of extruded aluminum provided with a tri-hollow cross-section enabling the frame to be reduced in weight without detracting from the strength of the frame.

3 Claims, 4 Drawing Figures

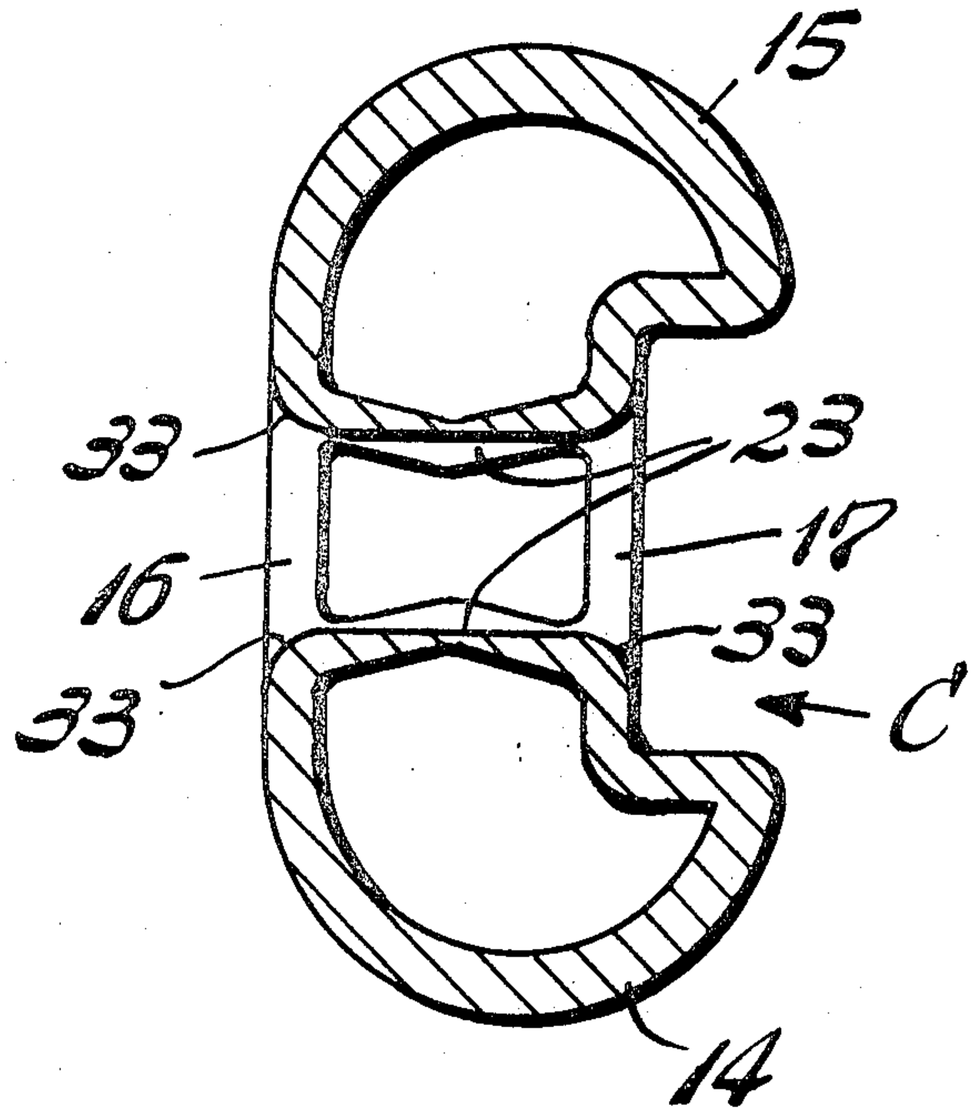


Fig. 1

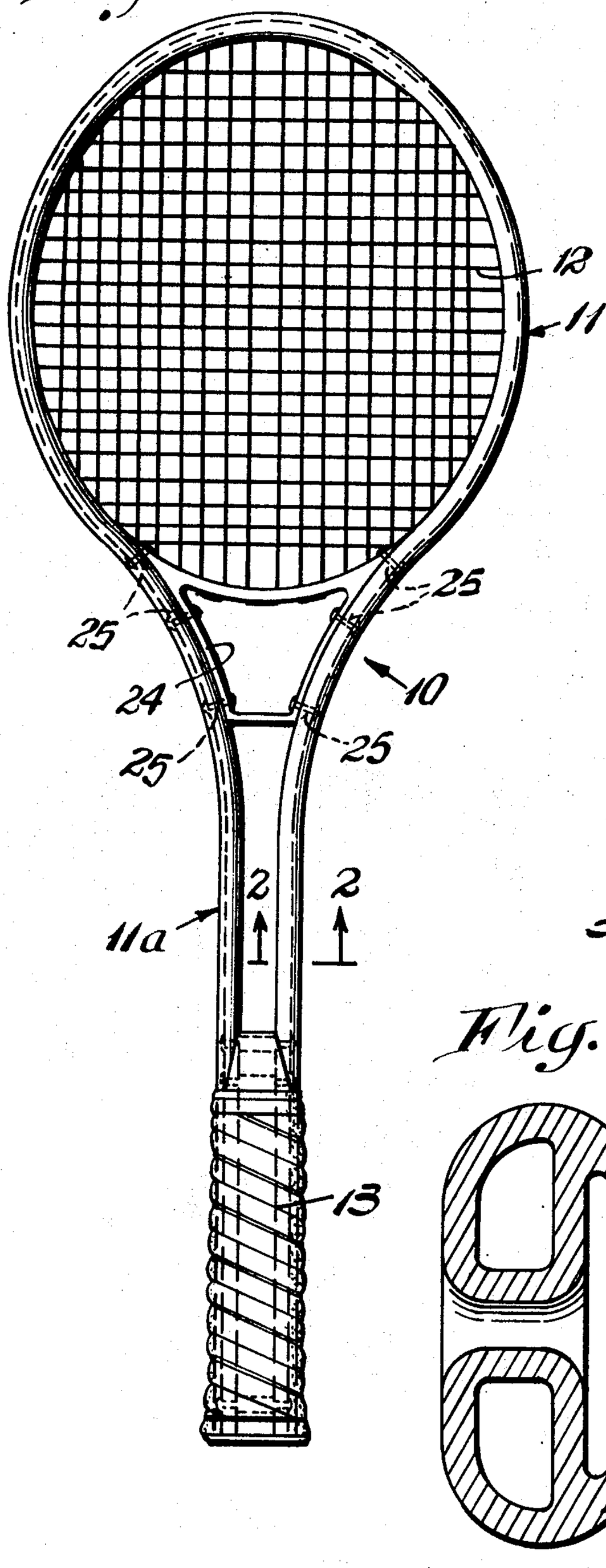


Fig. 2

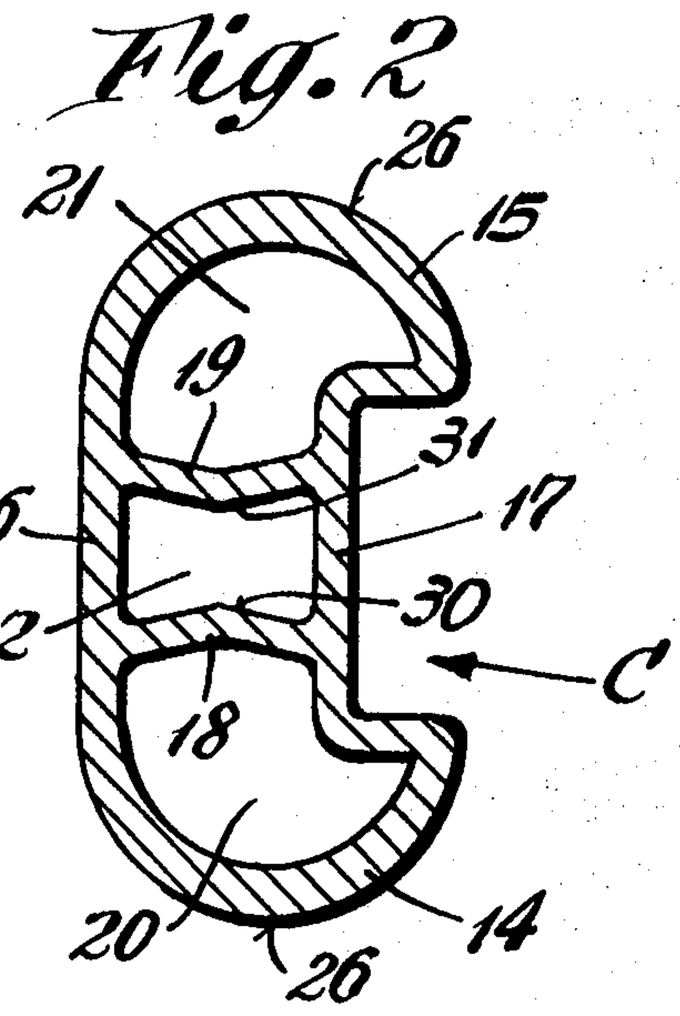


Fig. 3

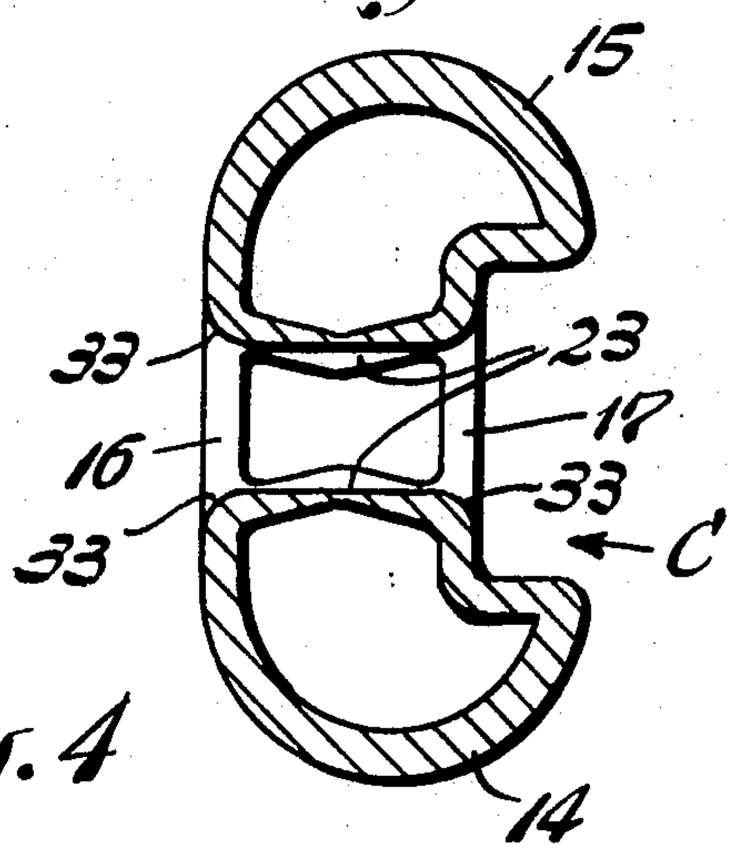
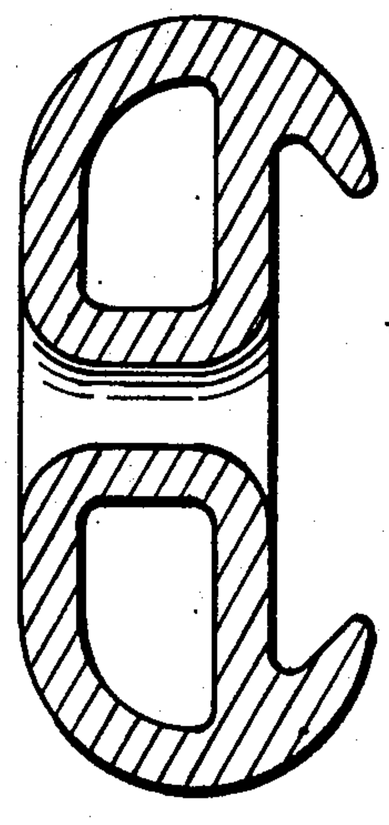


Fig. 4



PRIOR ART

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RACKET FRAME

Heretofore tennis racket frames have been formed from extruded aluminum units with tubular edge portions connected by a solid web in which string-receiving holes were formed. These frames had a tendency to be heavy in weight and resulted in a racket too heavy for play. Efforts to decrease the weight of the frame by reducing the wall thickness of the extrusion resulted in reducing the strength of the racket sufficiently to cause damage thereto during use. Also, difficulty was encountered in guiding the drill during the drilling of the stringing holes in the solid web and in bending the solid web smoothly during the formation of the bow. Other rackets have had solid metal frame sections and sheet metal frame sections but these have been difficult to produce in the required weight and strength relation.

The present invention overcomes these difficulties by providing an extruded aluminum frame unit which has a novel cross-section resulting in a frame which is lighter in weight while having the required strength to prevent damage to the frame in use, and also one which can be more readily bent and accurately drilled to provide the stringing holes therein.

This is accomplished by providing an extruded unit having tubular edge portions connected by spaced inner and outer webs, which webs are connected by a pair of longitudinal ribs disposed on opposite sides of the longitudinal axis of the unit, thus forming a central, substantially rigid thin-walled box section connecting the tubular edge portions. The extrusion is made as a flat strip and is thereafter bent to provide a string-receiving bow section and an integral handle-receiving section. Preferably, the outer web is recessed in the unit to provide a string-protecting channel around the bow and the string-receiving holes are drilled in the webs while the unit is in flat condition.

The accuracy of the drilling operation is enhanced, according to the present invention, by providing the opposite faces of the longitudinal ribs with drill centering and guiding surfaces which accurately control the drill as it forms the stringing holes.

A further advantage of the tri-hollow unit of the present invention resides in the fact that the wall thickness of the curved edge portions of the tubular edges can be made thicker than heretofore, while the walls of the webs and ribs can be made thinner.

Another advantage of the spaced webs connected by the longitudinal ribs has been found in that in bending the unit to form the bow, it more smoothly follows the desired curve rather than producing flat sections between the stringing holes as heretofore resulted.

Other features and advantages of the invention will be apparent from the specification and claims when considered in connection with the accompanying drawings in which:

FIG. 1 shows a plan view of a tennis racket having a frame of the present invention.

FIG. 2 is a sectional view through the extruded frame taken along lines 2—2 of FIG. 1.

FIG. 3 is a view similar to FIG. 2 taken through the bow section of the racket showing a stringing hole formed therein.

FIG. 4 is a section of the prior art racket (Type A).

As shown in the drawings, the tennis racket of the present invention comprises a frame formed by a unit 10 extruded as an elongate strip and bent to have a bow

section 11 shaped to receive theus usual strings 12 and continuations 11a of the frame extending down to receive a hand grip 13 to form the handle section for the racket.

While the unit can be formed of strong, lightweight metals and/or metal alloys or high strength plastics which are capable of being extruded, in the preferred form of the invention the unit 10 is formed as a flat strip of exturded aluminum or an alloy thereof having a cross-section, such as shown in FIG. 2, which has tubular edge sections 14, 15 integrally joined together by spaced inner and outer web sections 16, 17. As is shown in FIG. 2, the two web sections 16, 17 are joined by spaced longitudinal ribs 18, 19 integral therewith and disposed on opposite sides of the longitudinal axis of the extruded unit. The racket thus formed provides a tri-hollow frame unit having tubular edge sections 20, 21 connected by a substantially rigid hollow box section 22. If desired, the outer web 17 can be recessed in the unit to provide a straight-walled protecting channel C for the strings.

In accordance with the present invention the unit has a plurality of string-receiving holes 23 drilled in the webs in the bow section while the unit is in flat form. Thereafter the unit is bent to provide the required shape of the bow, as shown in FIG. 1, with the throat sections of the bow connected to a yoke 24 by rivets 25. The extending ends of the unit are in parallel relation, as shown in FIG. 1, and are connected to the suitable hand grip structure 13.

A feature of the invention resides in the fact that the unit, when extruded with the cross-section as shown in FIG. 2, is substantially lighter in weight than the extruded units heretofore used having a solid web connecting the tubular edge portions. Specifically, the unit was reduced in weight from 10.6 oz. minimum for prior units to a weight of about 9.7 oz. for the present construction.

This construction, while reducing the weight of the frame, permits the walls forming the curved surfaces 26 of the sections of the tubular edge portions to be reinforced and be thicker than the webs and the longitudinal ribs, as is clear from FIG. 2. By way of example, the thickness of the walls of the curve can be a minimum of 0.048 inch while the thickness of the webs and ribs are a minimum of 0.033 inch.

Further, the ribs 18, 19 connecting the two webs are shaped as shown in FIG. 2 with the opposed surfaces 30, 31 converging and forming centering and guiding surfaces for the drill (not shown) which provides the string-receiving holes in the bow. It will be seen from FIG. 3 that these surfaces will contact and guide the sides of the drill as it drills the holes 23, thus controlling the drill and preventing shifting of the drill and more accurate guiding and drilling of the holes. If desired, the edges of the holes are chamfered at 33 to prevent damage to the string.

It has also been found that with this construction, after the drilling operation and when the frame is bent into the bow shape, the two thin webs 16, 17 connected by the ribs 18, 19 having the angular relationship as shown in the drawings, will bend smoothly so that there is a constant curvature between the holes rather than the flat relationship between the holes which is present in the case of a frame having a solid web connecting the tubular portions.

It has been found that the tri-hollow frame of the present racket is not only lighter in weight than heretofore used, but that it is strengthened and will stand up under the stress of play as tested on a whacking machine (not shown) in which the rackets are mounted on two opposing arms rotated about a common axis by a motor at a rotational speed of 240 rpm with the distance of impact from the point of axis of rotation at 44 inches. The velocity at impact is 92 feet per second and the rackets are subjected to approximately 1,350 impacts per hour for a period of approximately 2½ hours. The maximum permissible set is six thirty-seconds inch under this test. It had been found that rackets of the present invention when subjected to this test showed no damage.

For example, a comparative test of five rackets of the prior art, such as shown in FIG. 4, having tubular edges connected with a solid web (Type A) and five rackets of the present invention (Type B) all having a frame weight of 10.6 oz. were tested until a permanent set of five thirty-seconds inch was recorded.

Test No.	Test Time	Type A Permanent Set (Net) RH (inches)	LH (inches)	Type B Permanent Set (Net) RH (inches)	LH (inches)
1.	30 min.	8/32	9/32	3/32	2/32 2/32
2.	30 min.	7/32	6/32	0	0
3.	30 min.	7/32	5/32	0	2/32
4.	45 min.	5/32	6/32	0	0
5.	2½ hrs.	6/32	5/32	1/32	1/32

Since all of the type B rackets were well within the acceptable maximum permanent set and passed the tests and all of the type A rackets exceeded the acceptable permanent set, it was clear that the racket of the present invention had a greater structural strength than the prior rackets of the same weight.

To show that lighter weight rackets of the present invention would have the required strength, four lesser weight rackets were tested on the whacking machine and successfully passed the test. For example:

Test No.	Frame Weight in Oz.	Time (hrs.)	Permanent Set (Net) RH (inches)	LH (inches)
6.	9.8	2½	4/32	1/32
7.	10	2½	5/32	4/32
8.	10.1	2½	3/32	4/32
9.	10.2	2½	4/32	3/32

The results of these tests show that the frames of the present invention could be reduced in weight by the desired amount and still have the required strength to maintain the permanent set at a low value to successfully pass the whacking test.

It will be seen, therefore, that the tri-hollow unit of

the present invention solves a long standing problem in the art of extruded metal frame rackets in that it permits the weight of the frame to be reduced so that playable frames of varying weights can be made and at the same time have the required strength to withstand the strains in use so as to provide an acceptable aluminum tennis racket.

Variations and modifications may be made within the scope of the claims and portions of the improvements may be used without others.

I claim:

1. A game racket frame including a string receiving bow section and an integral handle receiving section, said frame comprising

a hollow elongated seamless extruded unit, said unit having a cross section comprising

tubular edge sections,

spaced inner and outer webs connected between and integral with said tubular edge sections, and

a pair of spaced longitudinal ribs, each of said ribs being connected between said inner and outer webs, said longitudinal ribs being disposed on opposite sides of the longitudinal axis of said extruded unit so as to form a central hollow box and two hollow edge sections within said extruded unit;

the inner surface ofeach of said ribs having a geometrical configuration such that said inner surfaces converge toward each other so as to provide guiding and centering surfaces adapted to guide a drill for drilling string openings through said webs between said ribs.

2. The invention as defined in claim 1 wherein said outer web is recessed in the unit to provide a straight walled string protecting channel.

3. A game racket frame including a string receiving bow section and an integral handle receiving section, said frame comprising

a hollow elongated seamless extruded unit, said unit having a cross section comprising

tubular edge sections,

spaced inner and outer webs connected between and integral with said tubular edge sections, and

a pair of spaced longitudinal ribs, each of said ribs being connected between said inner and outer webs, said longitudinal ribs being disposed on opposite sides of the longitudinal axis of said extruded unit so as to form a central hollow box and two hollow edge sections within said extruded unit;

the portions of the unit forming an outer wall of the tubular edge sections being thicker than the webs and the longitudinal ribs.

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